

Solution Architecture

ComicCraft - AI Comic Story Creator using Gemini Models
Madras Institute of Technology | Department of Computer Technology (CT) | III Year, V Semester
Project period: 1 Oct 2026 - 2 Oct 2026

Browser: Jinja2 pages

FastAPI app (main.py, routes.py)

gemini_flash.py → gemini_pro.py → image_generator.py

layout_builder.py → exporters.py

static/panels | static/exports

Components

Component	Responsibility
app/main.py	Creates app, mounts /static, starts model warm-up thread
app/routes.py	Routes and pipeline orchestration, error handling
app/gemini_flash.py	Outline generation in JSON mode
app/gemini_pro.py	Story text in fixed Panel/Caption/Narration format
app/image_generator.py	Cached Stable Diffusion pipeline; saves PNG; returns URL
app/layout_builder.py	Regex-parses story; merges with images
app/exporters.py	FPDF export, one page per panel
templates/	index, comic_preview, export_success
static/	bg.jpg, fonts, panels, exports

Routes

Route	Method	Purpose
/	GET	Home form
/generate	POST	Run pipeline, show preview
/generate-comic/json	POST	JSON result
/export-success	GET	Download confirmation

/test-image	GET	Single test image
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External dependencies

- Google Gemini API over HTTPS (needs GEMINI_API_KEY)
- Hugging Face hub for first-time model download, then local cache